

Chapter 8: Graph Spaces & Spectral Clustering

One concept, one exercise per slide — practice as you go

Concept: PageRank — The Random Surfer Model

- Imagine a surfer randomly following links forever — PageRank is the long-run fraction of time spent on each page
- A page's rank is proportional to the sum of the ranks of pages linking to it, divided by their out-degree
- This is a fixed point: $\pi = M\pi$ — PageRank is the dominant eigenvector of the link matrix



Exercise: Page A is linked to by two pages: a high-authority page with few outgoing links, and a low-authority page with hundreds of outgoing links. Which incoming link contributes more to A's PageRank, and why?

Concept: The Damping Factor and Teleportation

- Pure link-following can get trapped in cycles or dead ends
- The Google Matrix adds teleportation: with probability $(1-d)$, jump to a uniformly random page
- Standard damping factor $d = 0.85$ — guarantees a unique, well-defined stationary ranking (Perron-Frobenius theorem)



Exercise: Why is the damping factor necessary — what could go wrong with PageRank on a real web graph if surfers ALWAYS followed links ($d=1$) and never teleported?

Concept: Dangling Nodes

- A dangling node has NO outgoing links — its column in the link matrix is all zeros
- Without special handling, probability mass simply vanishes at every dangling node
- Fix: redistribute a dangling node's rank uniformly across ALL nodes at every iteration



Exercise: After several PageRank iterations without dangling-node handling, you notice the rank values no longer sum to 1.0 (they're slowly shrinking). What is the most likely cause?

Concept: The Graph Laplacian

- $L = D - A$ (degree matrix minus adjacency matrix)
- Quadratic form: $f^T L f = (1/2) \sum_{ij} A_{ij} (f_i - f_j)^2$ — measures how 'non-smooth' f is over the graph
- L is always positive semidefinite — this quadratic form can never be negative
- The MULTIPLICITY of the zero eigenvalue equals the number of connected components



Exercise: A graph has 3 completely separate, disconnected clusters (no edges between them). How many zero eigenvalues will its Laplacian have?

Concept: The Fiedler Vector and RatioCut

- Finding the best way to CUT a graph into two balanced, weakly-connected pieces is NP-hard
- Relaxing the discrete cut problem to a continuous one gives a Rayleigh quotient minimization
- The solution is the Fiedler vector — the eigenvector for the SECOND-smallest Laplacian eigenvalue
- Sign of each entry in the Fiedler vector suggests which side of the cut that node belongs to



Exercise: Why do we use the SECOND-smallest eigenvalue (not the smallest) of the Laplacian to find a meaningful graph cut?

Concept: Spectral Clustering — The Full Pipeline

- 1. Build a similarity graph (e.g. k-nearest-neighbors) from raw feature data
- 2. Compute the (normalized) graph Laplacian
- 3. Extract the bottom-k eigenvectors — this becomes a new, k-dimensional embedding
- 4. Run ordinary k-means IN this new embedded space



Exercise: Why does spectral clustering run k-means on the LAPLACIAN EIGENVECTORS rather than on the original raw features?

Concept: Choosing k via the Spectral Gap

- Sort the Laplacian's eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
- A large GAP between λ_k and λ_{k+1} suggests k is a good number of clusters
- A vanishing gap suggests the data doesn't cleanly support that many clusters



Exercise: The sorted eigenvalues are: 0, 0.01, 0.02, 0.45, 0.9, 1.3, Where is the largest spectral gap, and what value of k does it suggest?

Concept: Common Mistake — Disconnected Graphs Break Spectral Clustering

- If the input graph has $c > 1$ connected components, the bottom c eigenvalues are ALL exactly zero
- The 'Fiedler vector' becomes ambiguous — any vector in that zero-eigenspace is equally valid
- Always check the number of near-zero eigenvalues before trusting the clustering result



Exercise: You run spectral clustering expecting $k=2$ clusters, but the graph actually has 2 completely disconnected components already. What goes wrong, and how would you detect it?

Mastery Check: Are You Ready to Move On?

If you can answer every question below confidently, you have mastered Chapter 8.

1. Explain the random-surfer interpretation of PageRank and why it is a dominant-eigenvector problem.
2. Explain why the damping factor and teleportation are necessary for PageRank to be well-defined.
3. Explain what a dangling node is and how its rank mass must be handled.
4. Write the Laplacian's quadratic form and explain what it measures about a function on a graph.
5. Explain why the number of zero eigenvalues of the Laplacian equals the number of connected components.
6. Explain why the Fiedler vector uses the SECOND-smallest eigenvalue, not the smallest.
7. Walk through all four steps of the spectral clustering pipeline from memory.
8. Explain how to choose k using the spectral gap, with a worked numeric example.
9. Diagnose why spectral clustering gives an ambiguous or wrong result on a disconnected graph.
10. Implement PageRank via power iteration on a small graph from scratch.